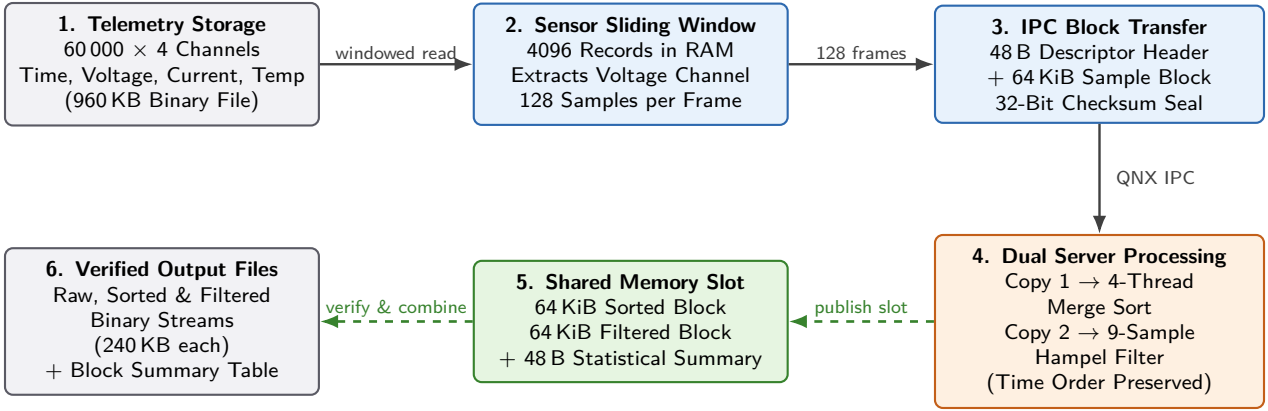

1. Telemetry Storage
60 000 × 4 Channels
Time, Voltage, Current, Temp
(960 KB Binary File)
windowed read
2. Sensor Sliding Window
4096 Records in RAM
Extracts Voltage Channel
128 Samples per Frame
128 frames
3. IPC Block Transfer
48 B Descriptor Header
+ 64 KiB Sample Block
32-Bit Checksum Seal
QNX IPC
4. Dual Server Processing
Copy 1 → 4-Thread
Merge Sort
Copy 2 → 9-Sample
Hampel Filter
(Time Order Preserved)
publish slot
5. Shared Memory Slot
64 KiB Sorted Block
64 KiB Filtered Block
+ 48 B Statistical Summary
verify & combine
6. Verified Output Files
Raw, Sorted & Filtered
Binary Streams
(240 KB each)
+ Block Summary Table